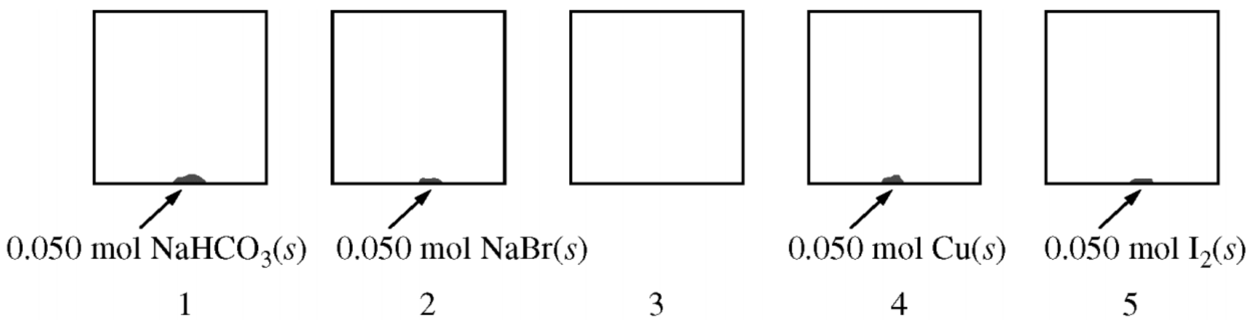
Chapter 06 Part 1: Ideal Gas Law

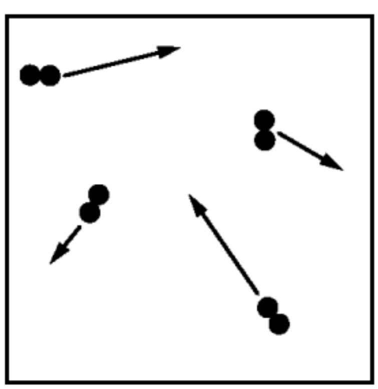
1. A 2 L container will hold about 4 g of which of the following gases at 0°C and 1 atm?

(A) SO_2
(B) N_2
(C) CO_2
(D) C_4H_8
(E) NH_3



At 27°C, five identical rigid 2.0 L vessels are filled with $N_2(g)$ and sealed. Four of the five vessels also contain a 0.050 mol sample of $NaHCO_3(s)$, $NaBr(s)$, $Cu(s)$, or $I_2(s)$, as shown in the diagram above. The volume taken up by the solids is negligible, and the initial pressure of $N_2(g)$ in each vessel is 720 mm Hg. All four vessels are heated to 127°C and allowed to reach a constant pressure.

2.



The gas particles in vessel 3 at 27°C are represented in the diagram above. The lengths of the arrows represent the speeds of the particles. Which of the following diagrams best represents the particles when vessel 3 is heated to 127°C?

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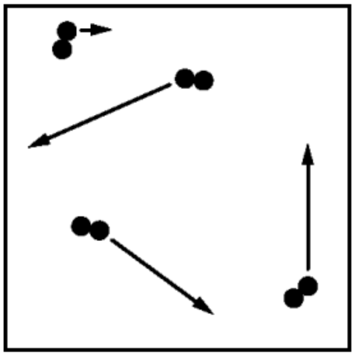
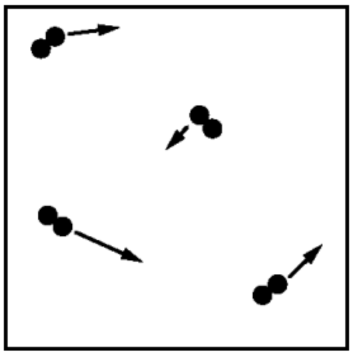
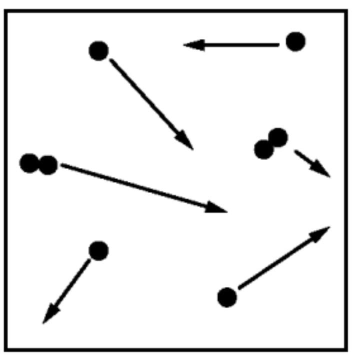
(A)



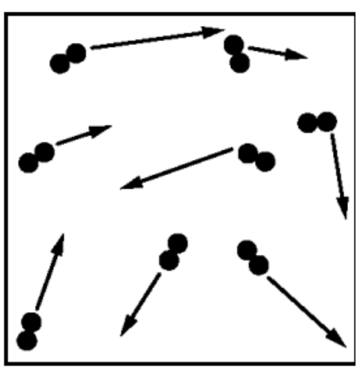
(B)



(C)



(D)



3. At 127°C, the pressure in vessel 1 is found to be higher than that in vessel 2. Which of the following reactions best accounts for the observation?

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(A) $NaHCO_3(s) \rightarrow Na(s) + HCO_3(s)$

(B) $NaHCO_3(s) \rightarrow NaH(s) + CO_3(s)$

(C) $2\ NaHCO_3(s) \rightarrow Na_2CO_3(s) + CO_2(g) + H_2O(g)$

(D) $2\ NaHCO_3(s) + N_2(g) \rightarrow 2\ NaNO_3(s) + C_2H_2(g)$

---

4. At standard temperature and pressure, a 0.50 mol sample of H_2 gas and a separate 1.0 mol sample of O_2 gas have the same

(A) average molecular kinetic energy

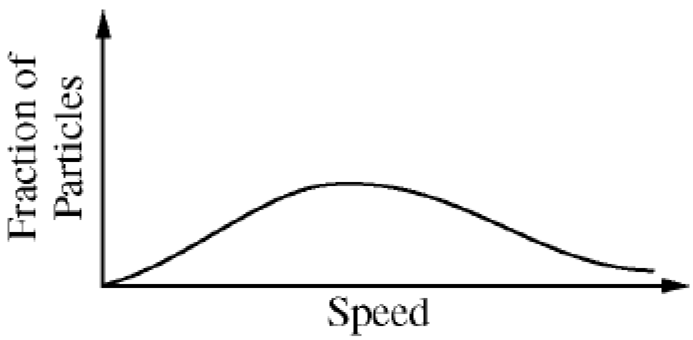
(B) average molecular speed

(C) volume

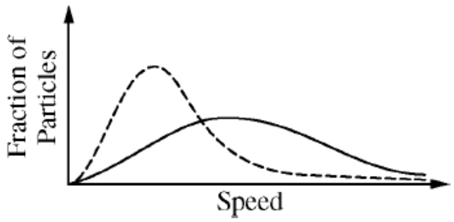
(D) effusion rate

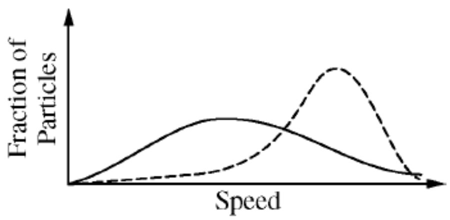
(E) density

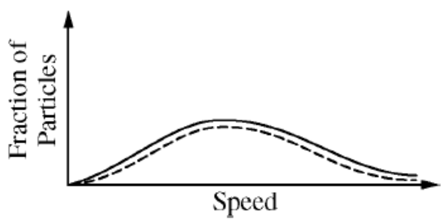
5.

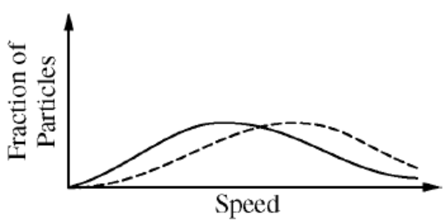


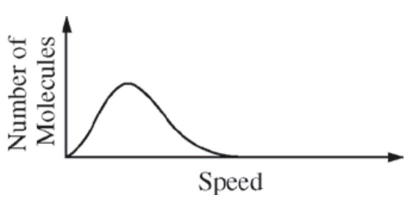
The distribution of speeds of $H_2(g)$ molecules at 273 K and 1 atm is shown in the diagram above. Which of the following best shows the speed distribution of $He(g)$ atoms under the same conditions of temperature and pressure?

(A) 

(B) 

(C) 

(D) 

6. 

The graph above shows the speed distribution of molecules in a sample of a gas at a certain temperature. Which of the following graphs shows the speed distribution of the same molecules at a lower temperature (as a dashed curve) ?

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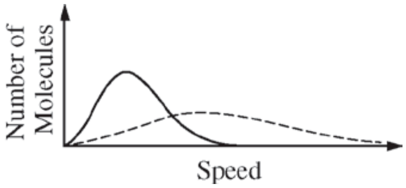
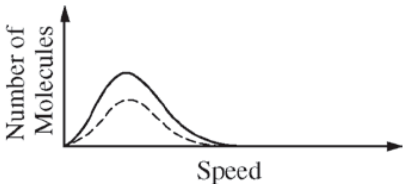
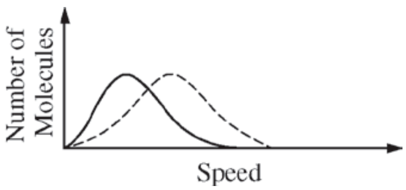
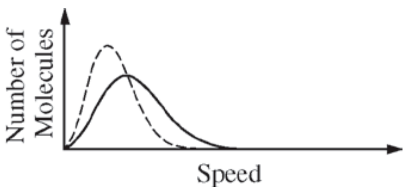
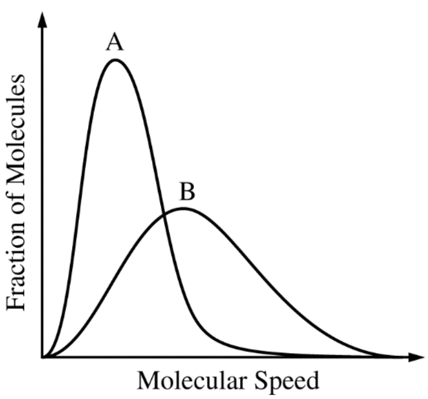
(A) 

(B) 

(C) 

(D) 

7.



The Maxwell-Boltzmann distributions of molecular speeds in samples of two different gases at the same temperature are shown above. Which gas has the greater molar mass?

(A) Gas A
(B) Gas B
(C) Both gases have the same molar mass.
(D) It cannot be determined unless the pressure of each sample is known.

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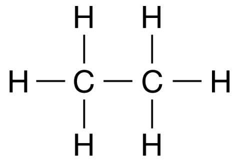
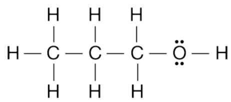
Compound	Molar Mass	Balanced Equation for the Combustion of One Mole of the Compound	$\Delta H^\circ{}_{comb}$ (kJ/mol_{rxn})
 Ethane	$30.\ \text{g/mol}$	$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2\,CO_2(g) + 3\,H_2O(l)$	-1560
 Propanol	$60.\ \text{g/mol}$	$C_3H_8O(l) + \frac{9}{2}O_2(g) \rightarrow 3\,CO_2(g) + 4\,H_2O(l)$	-2020
Unknown	?	$? +\ 5\,O_2(g) \rightarrow 4\,CO_2(g) + 4\,H_2O(l)$	-2240

8. A sealed 10.0 L flask at 400 K contains equimolar amounts of ethane and propanol in gaseous form. Which of the following statements concerning the average molecular speed of ethane and propanol is true?

(A) The average molecular speed of ethane is less than the average molecular speed of propanol.

(B) The average molecular speed of ethane is greater than the average molecular speed of propanol.

(C) The average molecular speed of ethane is equal to the average molecular speed of propanol.

(D) The average molecular speeds of ethane and propanol cannot be compared without knowing the total pressure of the gas mixture.

Refer to three gases in identical rigid containers under the conditions given in the table below.

Container	A	B	C
Gas	Methane	Ethane	Butane
Formula	CH_4	C_2H_6	C_4H_{10}
Molar mass (g/mol)	16	30.	58
Temperature (°C)	27	27	27
Pressure (atm)	2.0	4.0	2.0

9. The average kinetic energy of the gas molecules is

(A) greatest in container A

(B) greatest in container B

(C) greatest in container C

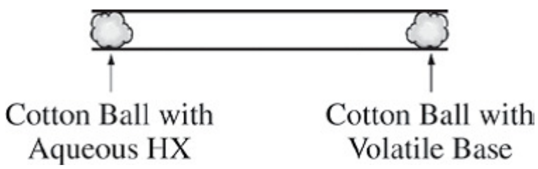
(D) the same in all three containers

10. The density of the gas, in g/L, is

Chapter 06 Part 1: Ideal Gas Law

(A) greatest in container A
(B) greatest in container B
(C) greatest in container C
(D) the same in all three containers

11.



The experimental apparatus represented above is used to demonstrate the rates at which gases diffuse. When the cotton balls are placed in the ends of a tube at the same time, the gases diffuse from each end and meet somewhere in between, where they react to form a white solid. Which of the following combinations will produce a solid closest to the center of the tube?

(A) HCl and CH_3NH_2
(B) HCl and NH_3
(C) HBr and CH_3NH_2
(D) HBr and NH_3

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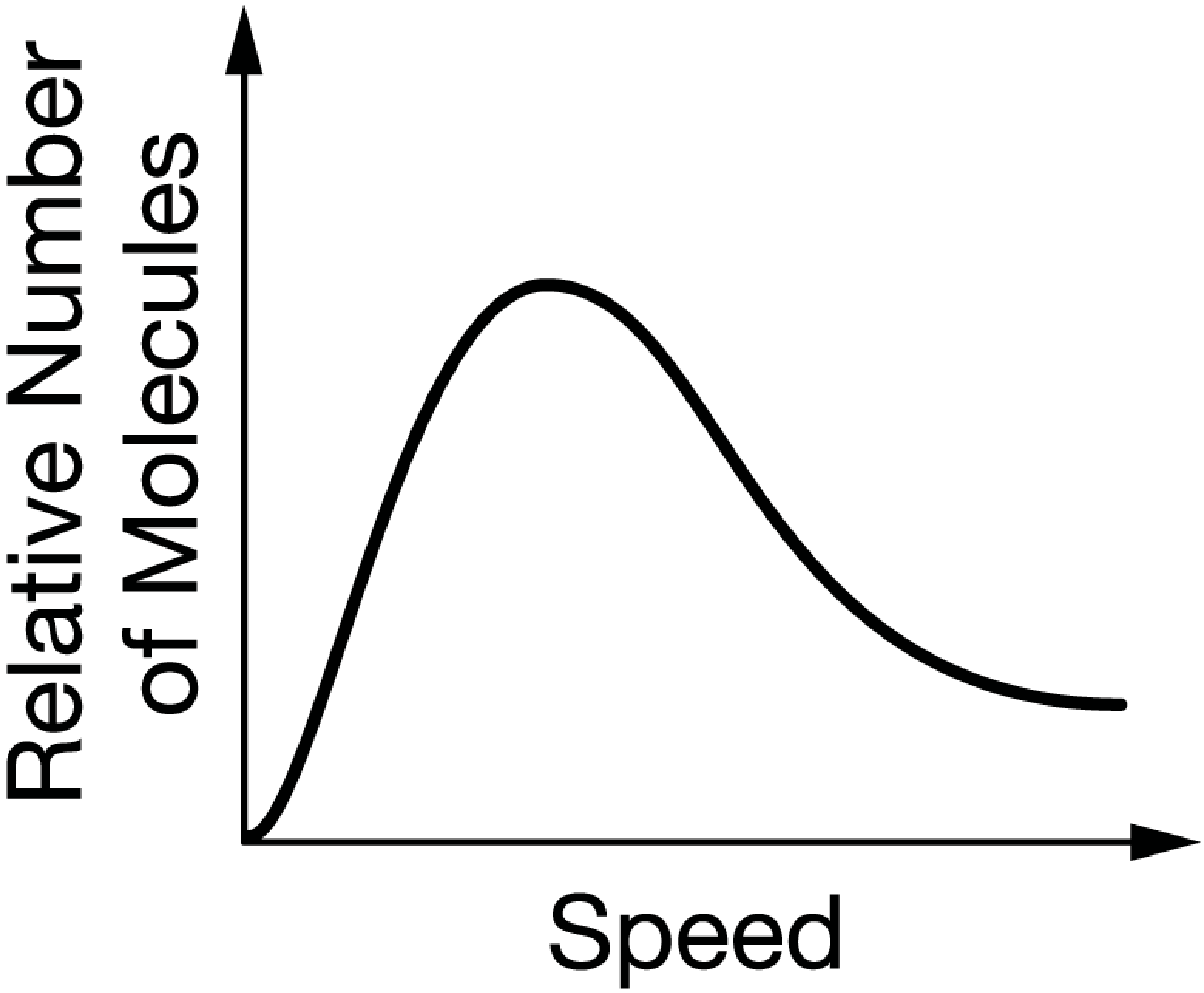
12.



The diagram above shows the distribution of speeds for a sample of $N_2(g)$ at $25^\circ C$. Which of the following graphs shows the distribution of speeds for a sample of $O_2(g)$ at $25^\circ C$ (dashed line) ?

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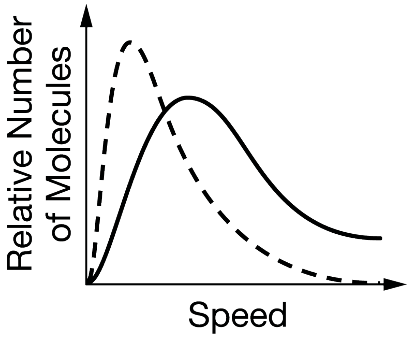
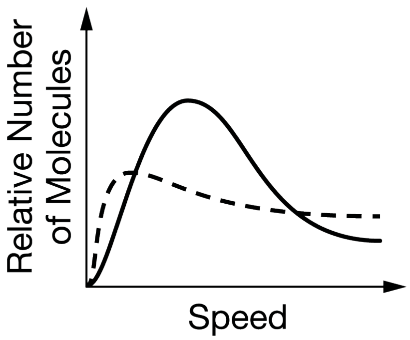
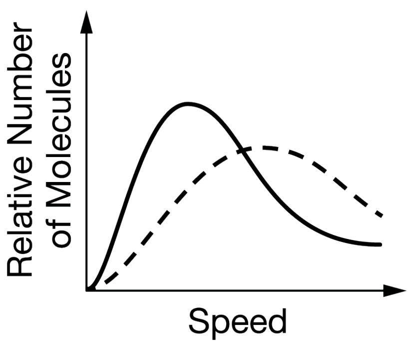
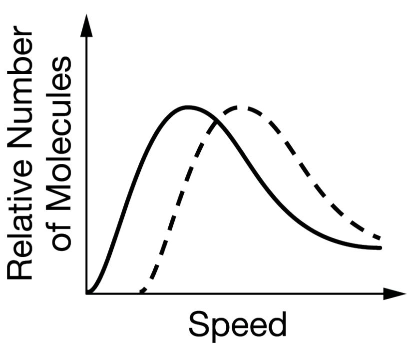
(A)


(B)


(C)


(D)


13. When a sample of oxygen gas in a closed container of constant volume is heated until its absolute temperature is doubled, which of the following is also doubled?

(A) The density of the gas

(B) The pressure of the gas

(C) The average velocity of the gas molecules

(D) The number of molecules per cm^3

(E) The potential energy of the molecules

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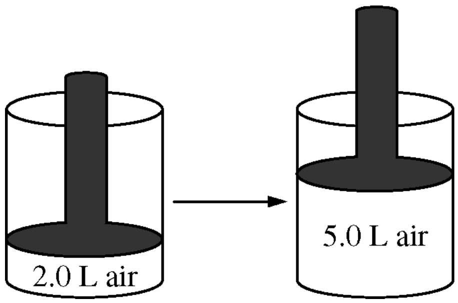
14.



The volume of a sample of air in a cylinder with a movable piston is 2.0 L at a pressure P_1, as shown in the diagram above. The volume is increased to 5.0 L as the temperature is held constant. The pressure of the air in the cylinder is now P_2. What effect do the volume and pressure changes have on the average kinetic energy of the molecules in the sample?

(A) The average kinetic energy increases.

(B) The average kinetic energy decreases.

(C) The average kinetic energy stays the same.

(D) It cannot be determined how the kinetic energy is affected without knowing P_1 and P_2.

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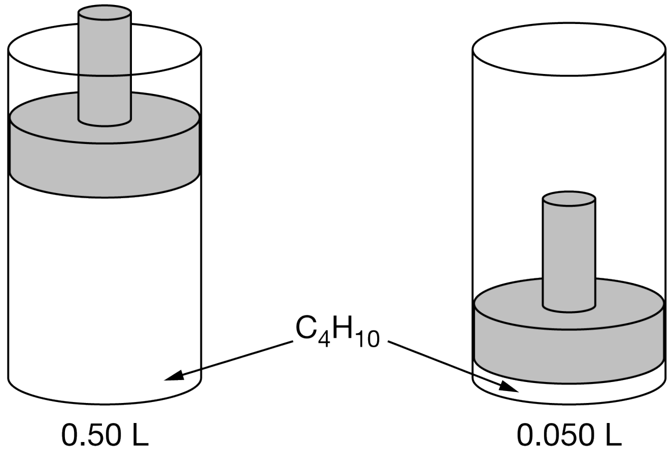
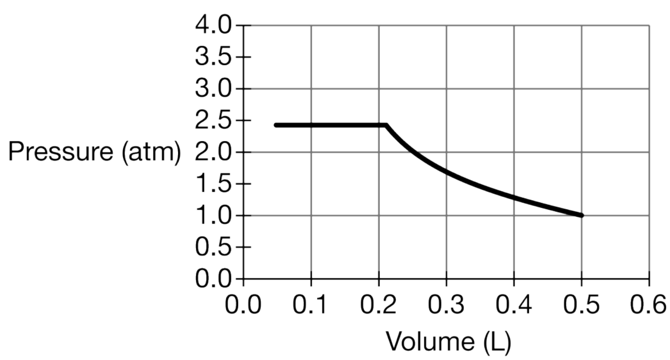


A sample of $C_4H_{10}(g)$ is placed in a rigid vessel with a movable piston. The volume of the gas is reduced by moving the piston downward while the temperature is maintained at $25^\circ C$, as shown above. The pressure of the gas versus its volume is plotted in the graph below.



15. Which of the following best explains the increase in pressure of the $C_4H_{10}(g)$ when its volume is decreased from 0.50 L to 0.21 L ?

(A) The molecules strike the walls of the vessel more frequently.

(B) The molecules strike the walls of the vessel with different orientations.

(C) The molecules strike the walls of the vessel with greater force.

(D) The molecules decompose to produce a greater number of particles.

16. Which of the following best accounts for the pressure in the vessel remaining constant as the volume decreases below 0.21 L?

(A) The density of the $C_4H_{10}(g)$ increases.

(B) The surface area of the interior of the vessel decreases.

(C) The temperature of the $C_4H_{10}(g)$ decreases.

(D) Some $C_4H_{10}(g)$ condenses to form $C_4H_{10}(l)$.

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17. When 4.0 L of He(*g*), 6.0 L of N_2(*g*), and 10. L of Ar(*g*), all at 0°C and 1.0 atm, are pumped into an evacuated 8.0 L rigid container, the final pressure in the container at 0°C is

(A) 0.5 atm
(B) 1.0 atm
(C) 2.5 atm
(D) 4.0 atm

18. When 6.0 L of He(*g*) and 10. L of N_2(*g*), both at 0^{o}C and 1.0 atm, are pumped into an evacuated 4.0 L rigid container, the final pressure in the container at 0^{o}C is

(A) 2.0 atm
(B) 4.0 atm
(C) 6.4 atm
(D) 8.8 atm
(E) 16 atm

19. A sample of an ideal gas in a sealed container occupies a volume of $50.0\ \mathrm{L}$ at $200\ \mathrm{K}$. The temperature of the gas is changed to $400\ \mathrm{K}$, and the pressure doubles. Which of the following must occur and why?

(A) Volume must decrease because the pressure is higher.
(B) Volume must increase because the temperature is higher.
(C) Volume must remain the same because the pressure and temperature change proportionately.
(D) Volume cannot be predicted without knowing the number of moles of gas in the sample.

The following questions refer to the below.

The table below contains information about samples of four different gases at 273 K. The samples are in four identical rigid containers numbered 1 through 4.

Container	Gas	Pressure (atm)	Mass of Sample (g)
1	He	2.00	?
2	Ne	2.00	?
3	?	2.00	16.0
4	SO_2	1.96	64.1

20. Under the conditions given, consider containers 1, 2, and 4 only. The average speed of the gas particles is

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(A) greatest in container 1
(B) greatest in container 2
(C) greatest in container 4
(D) the same in containers 1, 2, and 4

21. On the basis of the data provided above, the gas in container 3 could be

(A) CH_4
(B) O_2
(C) Ar
(D) CO_2

---

22. Two flexible containers for gases are at the same temperature and pressure. One holds 0.50 gram of hydrogen and the other holds 8.0 grams of oxygen. Which of the following statements regarding these gas samples is FALSE?

(A) The volume of the hydrogen container is the same as the volume of the oxygen container.
(B) The number of molecules in the hydrogen container is the same as the number of molecules in the oxygen container.
(C) The density of the hydrogen sample is less than that of the oxygen sample.
(D) The average kinetic energy of the hydrogen molecules is the same as the average kinetic energy of the oxygen molecules.
(E) The average speed of the hydrogen molecules is the same as the average speed of the oxygen molecules.

23. A 0.5 mol sample of He(*g*) and a 0.5 mol sample of Ne(*g*) are placed separately in two 10.0 L rigid containers at 25°C. Each container has a pinhole opening. Which of the gases, He(*g*) or Ne(*g*), will escape faster through the pinhole and why?

(A) He(*g*) will escape faster because the He(*g*) atoms are moving at a higher average speed than the Ne(*g*) atoms.
(B) Ne(*g*) will escape faster because its initial pressure in the container is higher.
(C) Ne(*g*) will escape faster because the Ne(*g*) atoms have a higher average kinetic energy than the He(*g*) atoms.
(D) Both gases will escape at the same rate because the atoms of both gases have the same average kinetic energy.

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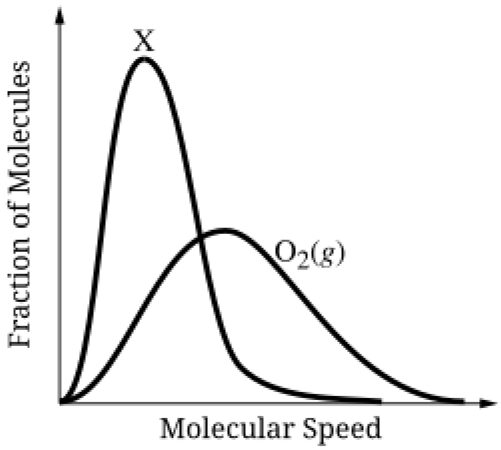
24.



Based on the Maxwell-Boltzmann distributions at $300\ \mathrm{K}$ shown above, which of the following gases could be represented by curve X?

(A) $Br_2(g)$

(B) $CH_4(g)$

(C) $He(g)$

(D) $N_2(g)$

25. A $1\ \mathrm{L}$ sample of helium gas at $25^\circ\mathrm{C}$ and $1\ \mathrm{atm}$ is combined with a $1\ \mathrm{L}$ sample of neon gas at $25^\circ\mathrm{C}$ and $1\ \mathrm{atm}$. The temperature is kept constant. Which of the following statements about combining the gases is correct?

(A) The average speed of the helium atoms increases when the gases are combined.

(B) The average speed of the neon atoms increases when the gases are combined.

(C) The average kinetic energy of the helium atoms increases when the gases are combined.

(D) The average kinetic energy of the helium atoms and neon atoms do not change when the gases are combined.

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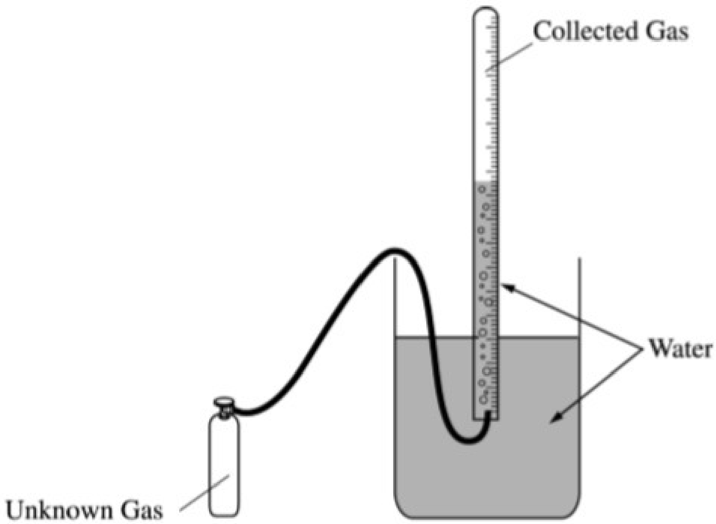
26.

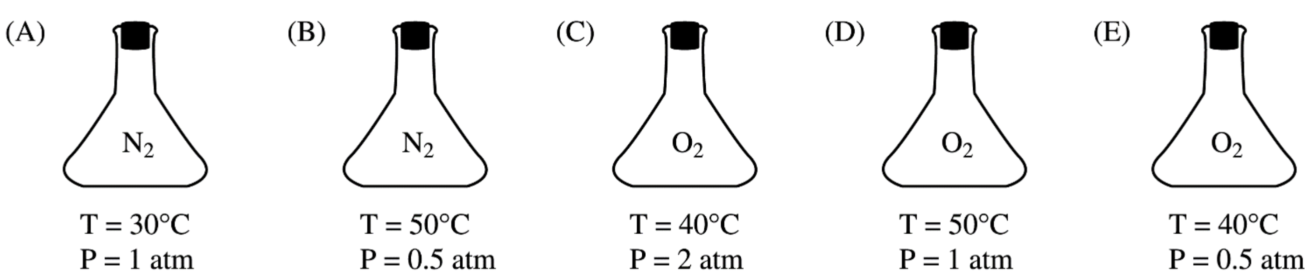


A sample of an unknown gas from a cylinder is collected over water in the apparatus shown above. After all the gas sample has been collected, the water levels inside and outside the gas collection tube are made the same. Measurements that must be made to calculate the molar mass of the gas include all of the following EXCEPT

(A) atmospheric pressure
(B) temperature of the water
(C) volume of gas in the gas-collection tube
(D) initial and final mass of the gas cylinder
(E) mass of the water in the apparatus

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits the statement. A choice may be used once, more than once, or not at all.

The following questions refer to the 1 L flasks shown below.



27. Which flask contains the sample with the greatest density?

(A) A
(B) B
(C) C
(D) D
(E) E

28. In which flask do the molecules have the greatest average speed?

(A) A
(B) B
(C) C
(D) D
(E) E

29. Which flask contains the smallest number of moles of gas?

(A) A
(B) B
(C) C
(D) D
(E) E

---

30. A gas mixture at $0^\circ C$ and $1.0\ atm$ contains $0.010\ mol$ of H_2, $0.015\ mol$ of O_2, and $0.025\ mol$ of N_2. Assuming ideal behavior, what is the partial pressure of hydrogen gas (H_2) in the mixture?

(A) About $0.010\ atm$, because there is $0.010\ mol$ of H_2 in the sample.
(B) About $0.050\ atm$, because there is $0.050\ mol$ of gases at $0^\circ C$ and $1.0\ atm$.
(C) About $0.20\ atm$, because H_2 comprises 20% of the total number of moles of gas.
(D) About $0.40\ atm$, because the mole ratio of H_2:O_2 : N_2 is $0.4 : 0.6 : 1$.

31. Equal masses of three different ideal gases, X, Y, and Z, are mixed in a sealed rigid container. If the temperature of the system remains constant, which of the following statements about the partial pressure of gas X is correct?

(A) It is equal to ⅓ the total pressure
(B) It depends on the intermolecular forces of attraction between molecules of X, Y, and Z.
(C) It depends on the relative molecular masses of X, Y, and Z.
(D) It depends on the average distance traveled between molecular collisions.
(E) It can be calculated with knowledge only of the volume of the container.

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32.

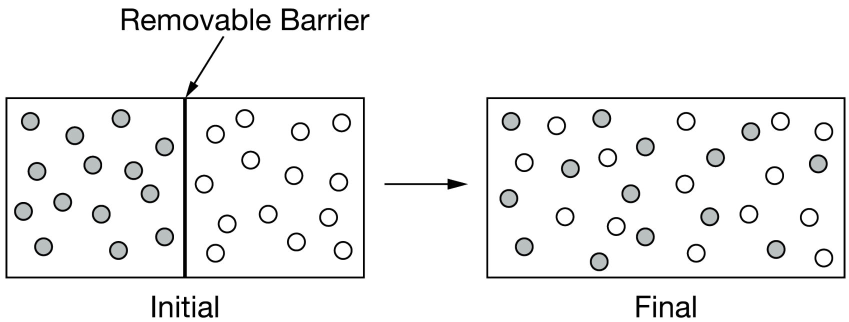


Diagram 1 above shows equimolar samples of two gases inside a container fitted with a removable barrier placed so that each gas occupies the same volume. The barrier is carefully removed as the temperature is held constant. Diagram 2 above shows the gases soon after the barrier is removed. Which statement describes the changes to the initial pressure of each gas and the final partial pressure of each gas in the mixture and also indicates the final total pressure?

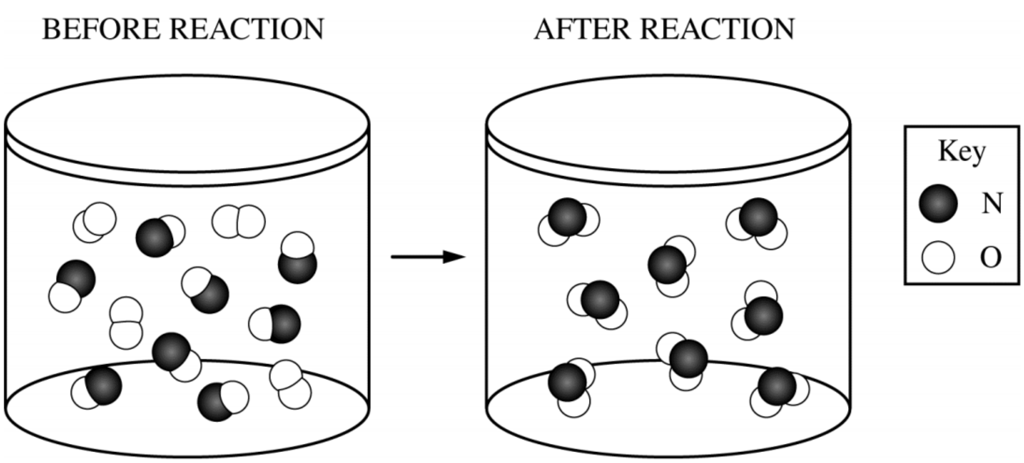
(A) The partial pressure of each gas in the mixture is double its initial pressure; the final total pressure is half the sum of the initial pressures of the two gases.

(B) The partial pressure of each gas in the mixture is double its initial pressure; the final total pressure is twice the sum of the initial pressures of the two gases.

(C) The partial pressure of each gas in the mixture is half its initial pressure; the final total pressure is half the sum of the initial pressures of the two gases.

(D) The partial pressure of each gas in the mixture is half its initial pressure; the final total pressure is the same as the sum of the initial pressures of the two gases.

33.



The reaction between NO(g) and O_2(g) to produce NO_2(g) in a rigid reaction vessel is represented in the diagram above. The pressure inside the container is recorded using a pressure gauge. Which of the following statements correctly predicts the change in pressure as the reaction goes to completion at constant temperature, and provides the correct explanation?

(A) The pressure will increase because the product molecules have a greater mass than either of the reactant molecules.

(B) The pressure will decrease because there are fewer molecules of product than of reactants.

(C) The pressure will decrease because the product molecules have a lower average speed than the reactant molecules.

(D) The pressure will not change because the total mass of the product molecules is the same as the total mass of the reactant molecules.

Refer to the following.

$PCl_5(g) \rightleftarrows PCl_3(g) + Cl_2(g)$

$PCl_5(g)$ decomposes into $PCl_3(g)$ and $Cl_2(g)$ according to the equation above. A pure sample of $PCl_5(g)$ is placed in a rigid, evacuated 1.00 L container. The initial pressure of the $PCl_5(g)$ is 1.00 atm. The temperature is held constant until the $PCl_5(g)$ reaches equilibrium with its decomposition products. The figures below show the initial and equilibrium conditions of the system.

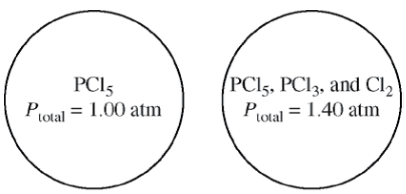


Figure 1: Initial Figure 2: Equilibrium

34. Which of the following is the most likely cause for the increase in pressure observed in the container as the reaction reaches equilibrium?

(A) A decrease in the strength of intermolecular attractions among molecules in the flask

(B) An increase in the strength of intermolecular attractions among molecules in the flask

(C) An increase in the number of molecules, which increases the frequency of collisions with the walls of the container

(D) An increase in the speed of the molecules that then collide with the walls of the container with greater force

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35. The pressure, in atm, exerted by 1.85 mol of an ideal gas placed in a 3.00 L container at 35.0°C is given by which of the following expressions?

(A) $\frac{(1.85)(0.0821)(308)}{3.00}$ atm

(B) $\frac{(1.85)(35.0)}{(0.0821)(3.00)}$ atm

(C) $\frac{3.00}{(1.85)(0.0821)(308)}$ atm

(D) $\frac{(1.85)(8.314)(308)}{3.00}$ atm

(E) $\frac{(3.00)(1.85)}{(0.0821)(35.0)}$ atm

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36. A sealed vessel contains 0.200 mol of oxygen gas, 0.100 mol of nitrogen gas, and 0.200 mol of argon gas. The total pressure of the gas mixture is 5.00 atm. The partial pressure of the argon is

(A) 0.200 atm
(B) 0.500 atm
(C) 1.00 atm
(D) 2.00 atm
(E) 5.00 atm

37. Equal masses of He and Ne are placed in a sealed container. What is the partial pressure of He if the total pressure in the container is 6 atm?

(A) 1 atm
(B) 2 atm
(C) 3 atm
(D) 4 atm
(E) 5 atm

38. A flask contains 0.25 mole of $SO_2(g)$, 0.50 mole of $CH_4(g)$, and 0.50 mole of $O_2(g)$. The total pressure of the gases in the flask is 800 mm Hg. What is the partial pressure of the $SO_2(g)$ in the flask?

(A) 800 mm Hg
(B) 600 mm Hg
(C) 250 mm Hg
(D) 200 mm Hg
(E) 160 mm Hg

39. Equal numbers of moles of He(g), Ar(g), and Ne(g) are placed in a glass vessel at room temperature. If the vessel has a pinhole-sized leak, which of the following will be true regarding the relative values of the partial pressures of the gases remaining in the vessel after some of the gas mixture has effused?

(A) $P_{He} < P_{Ne} < P_{Ar}$
(B) $P_{He} < P_{Ar} < P_{Ne}$
(C) $P_{Ne} < P_{Ar} < P_{He}$
(D) $P_{Ar} < P_{He} < P_{Ne}$
(E) $P_{He} = P_{Ar} = P_{Ne}$

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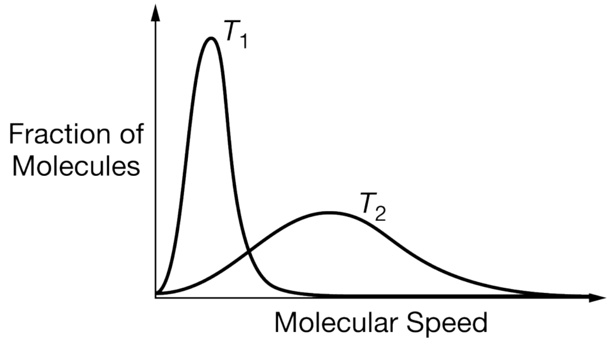
40.



The diagram above shows the distribution of molecular speed in a sample of gas at two different temperatures, T_1 and T_2. Which of the following is a correct interpretation of the information provided by the diagram?

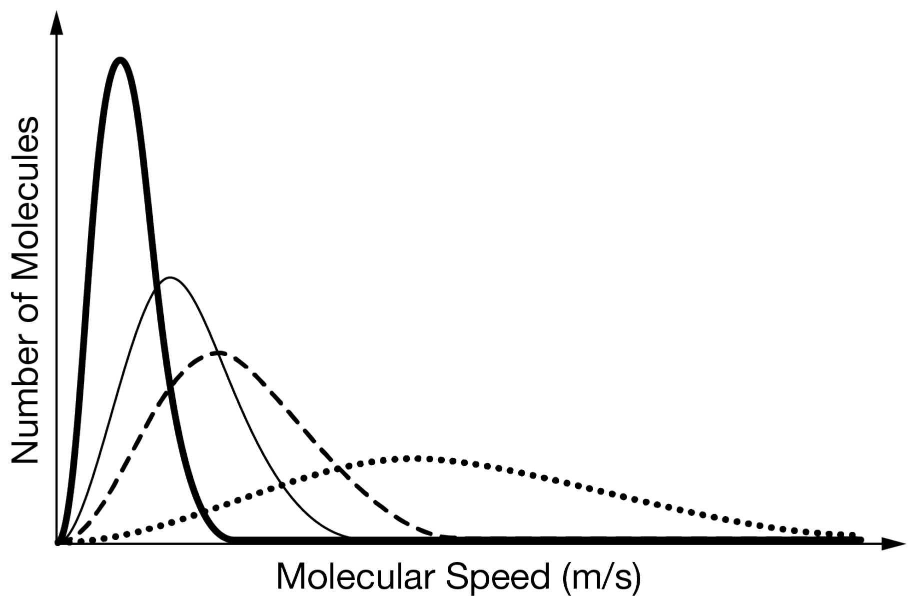
(A) The average speed of the molecules is greater at T_1 than at T_2.

(B) The taller peak corresponds to the higher temperature, T_1.

(C) There are fewer molecules of gas at T_2 than at T_1.

(D) The average kinetic energy of the molecules is greater at T_2 than at T_1.

41.



The graph above shows the distribution of molecular speeds for four different gases at the same temperature. What property of the different gases can be correctly ranked using information from the graph, and why?

(A) The densities of the gases, because as the density of a gas increases, the average speed of its molecules decreases.

(B) The pressures of the gases, because the pressure exerted by a gas depends on the average speed with which its molecules are moving.

(C) The volumes of the gases, because at a fixed temperature the volume of a gas can be calculated using the equation $PV = nRT$.

(D) The molecular masses of the gases, because the gas molecules have the same average kinetic energy and mass can be calculated using the equation $KE_{\text{avg}} = \frac{1}{2}mv^2$.

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42. A 2 L sample of $N_2(g)$ and a 1 L sample of $Ar(g)$, each originally at 1 atm and 0°C, are combined in a 1 L tank. If the temperature is held constant, what is the total pressure of the gases in the tank?

(A) 1 atm

(B) 2 atm

(C) 3 atm

(D) 4 atm

(E) 5 atm

43. An equimolar mixture of $N_2(g)$ and $Ar(g)$ is kept inside a rigid container at a constant temperature of 300 K. The initial partial pressure of Ar in the mixture is 0.75 atm. An additional amount of Ar was added to the container, enough to double the number of moles of Ar gas in the mixture. Assuming ideal behavior, what is the final pressure of the gas mixture after the addition of the Ar gas?

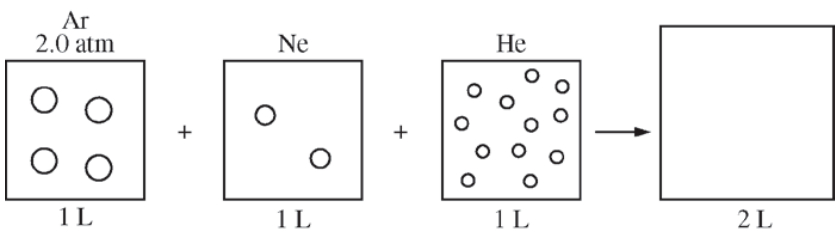
(A) 0.75 atm, because increasing the partial pressure of Ar decreases the partial pressure of N_2.

(B) 1.13 atm, because 33% of the moles of gas are N_2.

(C) 1.50 atm, because the number of moles of N_2 did not change.

(D) 2.25 atm, because doubling the number of moles of Ar doubles its partial pressure.

44.



The figure above represents three sealed 1.0 L vessels, each containing a different inert gas at 298 K. The pressure of Ar in the first vessel is 2.0 atm. The ratio of the numbers of Ar, Ne, and He atoms in the vessels is 2:1:6, respectively. After all the gases are combined in a previously evacuated 2.0 L vessel, what is the total pressure of the gases at 298 K?

(A) 3.0 atm

(B) 4.5 atm

(C) 9.0 atm

(D) 18 atm

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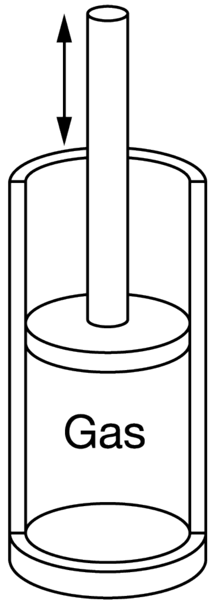
45.



A mixture of two gases, 0.01 mol of $C_4H_{10}(g)$ and 0.065 mol of $O_2(g)$, is pumped into a cylinder with a movable piston, as shown above. The mixture, originally at $200^{\circ}C$ and 1.0 atm, is sparked and the reaction represented below occurs.

$$2\,C_4H_{10}(g) + 13\,O_2(g) \rightarrow 8\,CO_2(g) + 10\,H_2O(g)$$

Which of the following is true after the product gases return to the original temperature and pressure, and why will the change occur? (Assume all gases behave ideally.)

(A) The piston will be higher than its original position, because the cylinder will contain a greater number of gas molecules.

(B) The position of the piston will be unchanged, because the total mass of the gases in the cylinder does not change.

(C) The position of the piston will be unchanged, because the temperature and pressure of the contents of the cylinder remain the same.

(D) The piston will be lower than its original position, because the product molecules are smaller than the reactant molecules.

46. A balloon filled with 0.25 mol of He(g) at 273 K and 1 atm is allowed to rise through the atmosphere. Which of the following explains what happens to the volume of the balloon as it rises from ground level to an altitude where the air temperature is 220 K and the air pressure is 0.1 atm?

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(A) The volume will increase because the decrease in air pressure will have a greater effect than the decrease in temperature.

(B) The volume will remain unchanged because of the counteracting effects of the decrease in temperature and the decrease in air pressure.

(C) The volume will decrease because the decrease in temperature will have a greater effect than the decrease in air pressure.

(D) It cannot be determined whether the volume of the balloon will increase, decrease, or remain the same without knowing the initial volume of the balloon.

47. A rigid metal tank contains oxygen gas. Which of the following applies to the gas in the tank when additional oxygen is added at constant temperature?

(A) The volume of the gas increases.

(B) The pressure of the gas decreases.

(C) The average speed of the gas molecules remains the same.

(D) The total number of gas molecules remains the same.

(E) The average distance between the gas molecules increases.

48. A sample of an ideal gas is cooled from $50.0^{o}C$ to $25.0^{o}C$ in a sealed container of constant volume. Which of the following values for the gas will decrease?

I. The average molecular mass of the gas

II. The average distance between the molecules

III. The average speed of the molecules

(A) I only

(B) II only

(C) III only

(D) I and III

(E) II and III

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49. A student was assigned the task of determining the molar mass of an unknown gas. The student measured the mass of a sealed 843 mL rigid flask that contained dry air. The student then flushed the flask with the unknown gas, resealed it, and measured the mass again. Both the air and the unknown gas were at 23.0°C and 750. torr. The data for the experiment are shown in the table below.

Volume of sealed flask	843 mL
Mass of sealed flask and dry air	157.70 g
Mass of sealed flask and unknown gas	158.08 g

a. Calculate the mass, in grams, of the dry air that was in the sealed flask. (The density of dry air is 1.18 g L^{-1} at 23.0°C and 750. torr.)

b. Calculate the mass, in grams, of the sealed flask itself (i.e., if it had no air in it).

c. Calculate the mass, in grams, of the unknown gas that was added to the sealed flask.

d. Using the information above, calculate the value of the molar mass of the unknown gas.

After the experiment was completed, the instructor informed the student that the unknown gas was carbon dioxide (44.0 g mol^{-1}) .

e. Calculate the percent error in the value of the molar mass calculated in part (d).

f. For each of the following two possible occurrences, indicate whether it by itself could have been responsible for the error in the student's experimental result. You need not include any calculations with your answer. For each of the possible occurrences, justify your answer.

Occurrence 1: The flask was incompletely flushed with $CO_2(g)$, resulting in some dry air remaining in the flask.

Occurrence 2: The temperature of the air was 23.0°C, but the temperature of the $CO_2(g)$ was lower than the reported 23.0°C.

g. Describe the steps of a laboratory method that the student could use to verify that the volume of the rigid flask is 843 mL at 23.0°C. You need not include any calculations with your answer.

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50. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

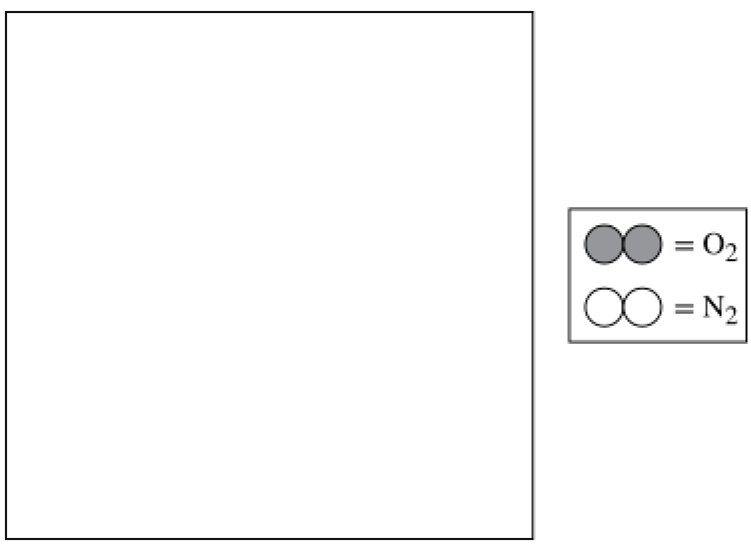
Dry air is comprised of N_2, O_2, and traces of other gases.

A student obtains a $2.00\ \mathrm{L}$ container filled with a sample of dry air at a total pressure of $1.30\ \mathrm{atm}$ and a temperature of $295\ \mathrm{K}$.

(a) Calculate the total number of moles of gas in the container.

(b) The student cools the gaseous mixture until either N_2 or O_2 condenses. Which gas will condense first? Justify your answer in terms of the types and relative strengths of the intermolecular forces in N_2 and O_2.

(c) Using the legend given, draw at least three molecules of each type of gas in the sealed container represented below when one of the substances has condensed, but the other substance is still a gas.



(d) In this sample, 78% of the molecules are N_2 and 21% of the molecules are O_2. Calculate the total number of grams of the substance that condenses first.